

Customer Shopping Behavior Analysis

1. Project Overview

The Customer Behavior Dashboard is an interactive data analytics project developed using SQL, Python, and Power BI to analyze customer purchasing patterns and business performance. The dashboard provides insights into customer demographics, subscription status, revenue, and sales across different product categories and age groups.

2. Dataset Summary

- Rows – 3,900
- Columns – 18
- Key Features:
 - Customer demographics (Age , Gender, Subscription Status, Location)
 - Purchase details (Category , Item purchased, Purchase amount , Season, Size , Color)
 - Shopping behavior (Discount applied , Promo code used , Previous purchase , Review rating , Shipping type , Frequency of purchases)
- Missing Data : 37 values in review rating

3.Exploratory Data Analysis using Python

- Data Loading: Data loaded using [pandas](#).
- Initial Exploration: Used [df.info \(\)](#) to check structure and [.describe\(\)](#) for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Pu
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	
5	6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standard	Yes	Yes	
6	7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipping	Yes	Yes	
7	8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes	Free Shipping	Yes	Yes	
8	9	26	Male	Coat	Outerwear	97	West Virginia	L	Silver	Summer	2.6	Yes	Express	Yes	Yes	
9	10	57	Male	Handbag	Accessories	31	Missouri	M	Pink	Spring	4.8	Yes	2-Day Shipping	Yes	Yes	

```

Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season          0
Review Rating    0
Subscription Status  0
Shipping Type    0
Discount Applied 0
Promo Code Used  0
Previous Purchases 0
Payment Method   0
Frequency of Purchases 0
dtype: int64

```

- **Missing Data Handling:** Check for null values and imputed values in the review rating column .
- **Column Standardization:** Rename columns snake case for better readability and documentattion.
- **Feature Engineering:**
 - Created age group column by binning customer ages.
- **Data Consistency check :** verified discount_applied and promo_code_used were redundant dropped promo_code_used.
- **Database Integration :** Connected python script to postgresql and loaded the cleaned dataframe into the database for sql analysis.

4.Data Analysis using sql

1. Revenue by gender:Compared total revenue generated by male vs female customers.

	gender 	revenue 
1	Female	75191
2	Male	157890

2.High spending discount users : identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
15	35	91
16	37	69
17	40	60

3. Top 5 Products by rating: found products with highest average review rating,

	item_purchased text	average product rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.81
4	Hat	3.81
5	T-shirt	3.78

4.Shipping type comparision :

Compared average purchases amount between Standard and Express.

	shipping_type text	round numeric
1	Standard	58.46

5.Total reveue by Subscription status :

Found total revenue by subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6.Discount rate on item_purchased:

Got discount_rate based on item purchased.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

7.Comparision on Customer segement :

Classified customers as loyal , new and returning.

	customer_segment text	number of customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

5. Dashboard



6. Bussiness Recommendation

- 1. Boost subscription: provide exclusive benefits for subscribers.
- 2. Customer loyalty progress :Reward repeater buyers to more then into the ‘Loyal’ segment.
- 3. Retrive Discount policy :balance sales boosts with margin control.